

0.1 Section 2.3 - Functions

Definition 1

Let A and B be sets

A **relation** from A to B is a subset of $A \times B$

Example 1

Let $A = \{1, 4, 9\}$ and $B = \{\pm 1, \pm 2, \pm 3\}$

1. $|A \times B| = 3 \cdot 6 = 18$
2. Let R be the relation from A to B defined by

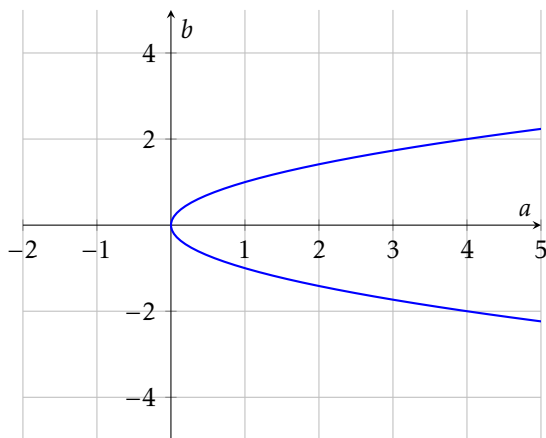
$$R = \{(a, b) \mid a \in A, b \in B, a = b^2\}$$

$$R = \{(1, 1), (1, -1), (4, 2), (4, -2), (9, 3), (9, -3)\}$$

Note

R is a relation but not a function

Why?



This graph of the relation fails the vertical line test, hence it is not a function

There exists an element of A that corresponds to more than one element of B

a	b
1	-1
1	1
4	-2
4	2
9	-3
9	3

Every function is a relation but not all relations are functions

Definition 2 (Function)

Let A and B be sets

A **function** f from A to B , denoted as

$$f : A \rightarrow B$$

is a relation from A to B such that

$$\forall a \in A \exists! b \in B, f(a) = b$$

$\exists!$ means "there exists a **unique**..."

Definition 3 (Domain and Codomain)

We call the set A the **domain** of f and the set B the **codomain**.

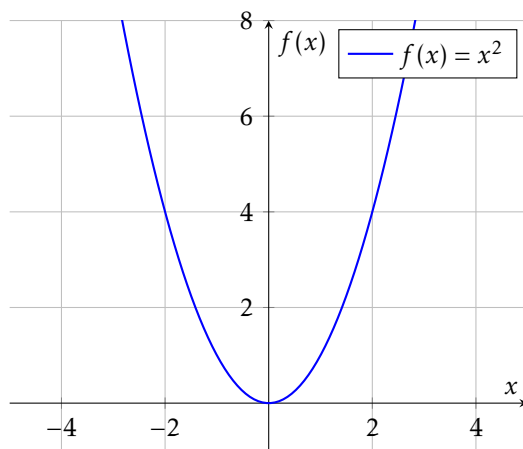
The **range** or **image** of f is the set

$$\text{Im}(f) = \{b \in B \mid \exists a \in A, f(a) = b\}$$

Example 2

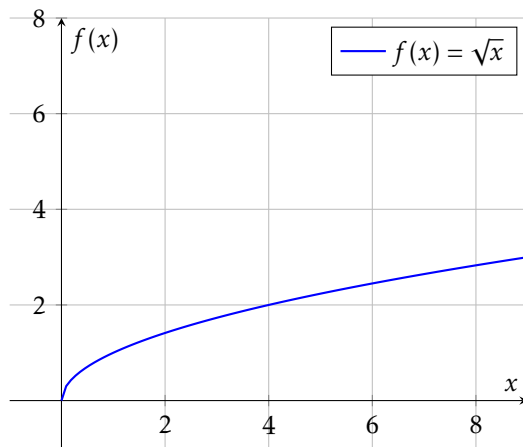
Determine if the following are functions

1. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$



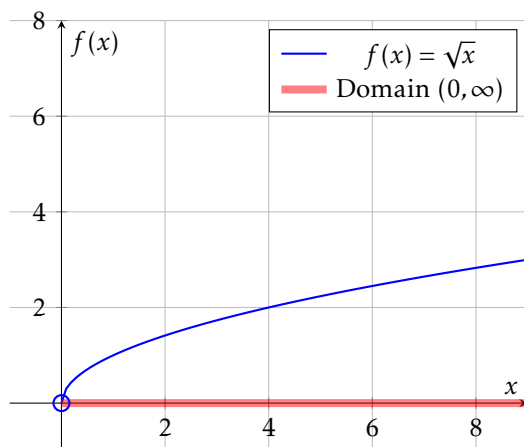
Domain: x axis
 Codomain: y axis
 $\text{Im}(f) = [0, \infty]$
 $\therefore$ Yes

2. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sqrt{x}$



No since not all inputs are valid

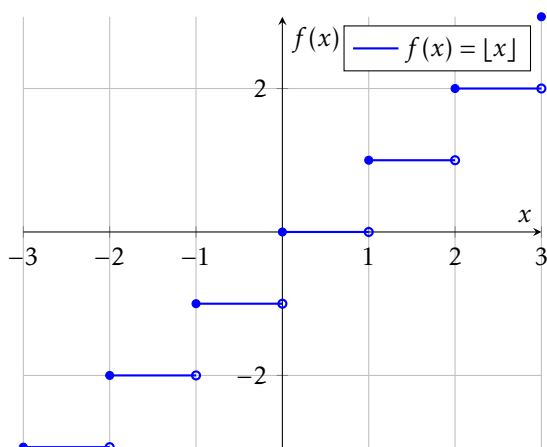
3. $f : \mathbb{R}^+ \rightarrow \mathbb{R}, f(x) = \sqrt{x}$



$$\text{Im}(f) = (0, \infty)$$

Yes since all the inputs have a unique output (the origin is excluded since the domain is $\mathbb{R}^+$)

4. $f : \mathbb{R} \rightarrow \mathbb{Z}, f(x) = \lfloor x \rfloor$ (floor function, aka greatest integer function)

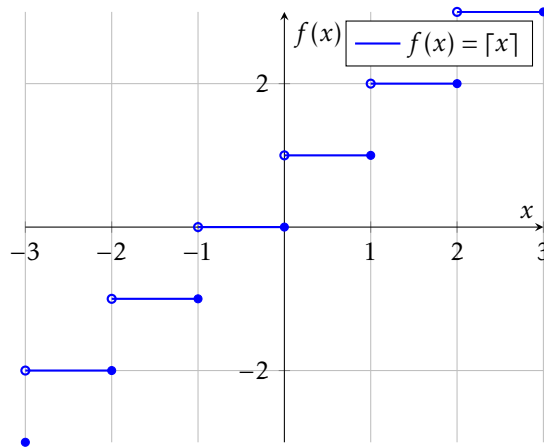


x	$\lfloor x \rfloor$
-1	-1
0	0
0.5	0
0.99	0
1.01	1

$$\text{Im}(f) = \mathbb{Z}$$

All of x axis is domain, but only **integer** parts of y axis is codomain

5. $f : \mathbb{R} \rightarrow \mathbb{Z}$, $f(x) = \lceil x \rceil$ (ceiling function, aka least integer function)



x	$\lceil x \rceil$
-1	-1
0	0
0.5	1
0.99	1
1.01	2

$$\text{Im}(f) = \mathbb{Z}$$

All of x axis is domain, but only **integer** parts of y axis is codomain

Definition 4 (Surjective)

Let $f : A \rightarrow B$ be a function in the set A and B

We say f is **onto** or **surjective** if

$$\text{Im}(f) = B$$

or

$$\forall b \in B \exists a \in A, f(a) = b$$

or

All of codomain is covered by the function

Definition 5 (Injective)

We say f is **one to one** or **injective** if

$\forall b \in B$, the equation $f(a) = b$ has at most one solution

Another way of thinking of this is:

Injective if it passes the horizontal line test

or

there is a unique b for all a

or

no two inputs have the same output

or

$$f(a) = f(b) \implies a = b$$

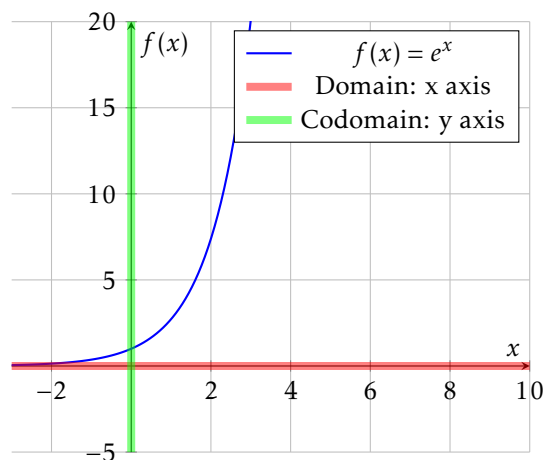
Definition 6 (Bijective)

If f is both **surjective** and **injective**, then we say f is **bijective**

Example 3

Determine if the following functions are injective, surjective, and / or bijective

1. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^x$

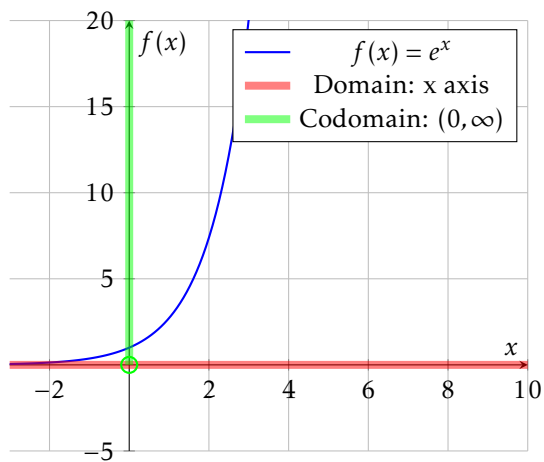


Injective since all x values have a valid output

Not surjective since not all values on the y axis have a corresponding input value

Not bijective since it's not surjective

2. $f : \mathbb{R} \rightarrow \mathbb{R}^+, f(x) = e^x$



Injective since all x values have a valid output

Surjective since all values on the y axis have a corresponding input value

Bijective since it's both surjective and injective

There's more examples, write them down when you get the chance

Definition 7 (Composition)

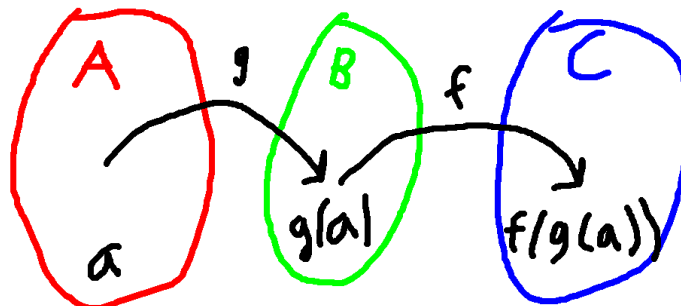
Given the functions $f : B \rightarrow C$ and $g : A \rightarrow B$

Where A, B, C are sets, then we define

$$f \circ g : A \rightarrow C \text{ by}$$

$$(f \circ g)(a) = f(g(a))$$

called the **composition** of f and g



Example 4 (Examples of Compositions)

Let $f(x) = x - 3$, $g(x) = x^3 + 9x^2 + 27x + 27 = (x + 3)^3$

Compute the following

1. $(f \circ g)(x)$

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= (x^3 + 9x^2 + 27x + 27) - 3 \\ &= x^3 + 9x^2 + 27x + 24\end{aligned}$$

2. $(g \circ f)(x)$

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= (x - 3)^3 + 9(x - 3)^2 + 27(x - 3) + 27 \\ &= [(x - 3) + 3]^3 \\ &= x^3\end{aligned}$$